GGC-H-00110(GSA-FSS)
February 10, 1964

INTERIM FEDERAL SPECIFICATION

HANDLING TOOL, SHEET MATERIAL (VACUUM TYPE)

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 This specification covers the requirements for one type of handling tool used for handling sheet material.

1.1.1 Federal specification coverage. - Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. - The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Folding, Fiberboard, Corrugated, Triple Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing).

Federal Standards:
Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

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Military Specifications:
MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.
MIL-B-43014 - Boxes, Water Resistant Paperboard; Folding, Set-Up.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustration. - The illustration shown herein is for the convenience of identification and is not intended to preclude the purchase of handling tools which are otherwise in accordance with this specification.

3.2 Materials. - The materials shall be as specified herein. Materials not definitely specified shall be of the quality best suited for the purpose intended.

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3.2.1 Aluminum. - The aluminum component of the handling tool shall be standard commercial quality.

3.2.2 Synthetic rubber. - Rubber components of the tool shall be fabricated from oil resisting neoprene.

3.3 Construction. - The handling tool shall be of sturdy construction so as to withstand hard and rough usage. Construction shall conform to the requirements of this specification and shall withstand the tests specified herein.

3.4 Design. - The handling tool shall be similar to figure 1 and shall consist of a handle, tw vacuum cups, and a vacuum release mechanism.

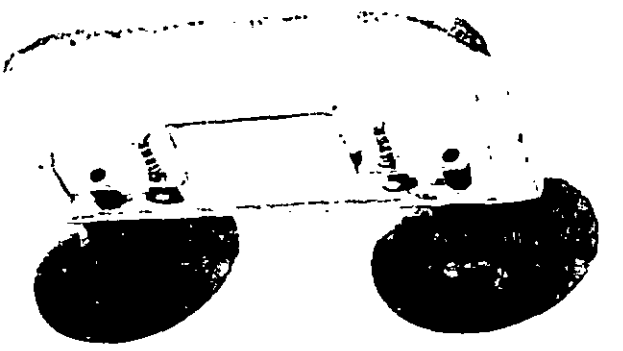


Figure 1. - Handling tool, sheet material (vacuum type).

3.4.1 Dimensions. - The normal overall dimensions of the handling tool shall be 10-1/2 inche long, 5-3/4 inches high, and 4 inches wide.

3.5 Performance. - The handling tool shall be capable of withstanding a pull of 120 pounds wi attached, by vacuum, to a smooth surface.

3.6 Details of components.

3.6.1 Handle. - The handle shall be fabricated from aluminum, properly formed to fit the han comfortably, and finished smooth and free of burrs or sharp projections.

3.6.2 Vacuum cups. - The vacuum cups shall be fabricated from synthetic rubber (neoprene) wh is properly compounded for strength, resilience, and flexibility. The vacuum cups shall be 4 inch in diameter. The tolerance shall be plus or minus 1/4 inch.

3.6.3 Vacuum release mechanism. - The vacuum release mechanism shall be suitably located to prevent accidental release and afford convenient operation. The mechanism shall be spring loaded to maintain a positive vacuum seal until intentionally released.

3.7 Weight. - The handling tool shall weigh not more than 1-1/2 pounds.

3.8 Workmanship. - Workmanship shall be in accordance with high grade industrial practice.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specifie herein. Except as otherwise specified, the supplier may utilize his own facilities or any commer laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling.

4.2.1 Inspection lot. - All handling tools manufactured by a single production process and offered for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.2.2 Sampling for visual and dimensional inspection. - A random sample shall be selected from each lot in accordance with Standard MIL-STD-105 at inspection level II. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.2.3 Sampling for acceptance tests. - A random sample shall be selected from each lot in accordance with Standard MIL-STD-105 at inspection level S-2. The AQL shall be 6.5 percent defective.

4.3 Visual and dimensional inspection. - Each of the sample handling tools selected in accordance with 4.2.2 shall be visually and dimensionally examined to determine compliance with this specification. Any sample containing one or more defects shall be rejected.

4.4 Acceptance tests. - Each sample selected in accordance with 4.2.3 shall be tested in accordance with 4.5 to verify compliance with this specification. Any sample which does not meet the requirements for any of these tests shall be rejected.

4.5 Oil resistance test. - Each sample handling tool shall be completely submerged in hydraulic fluid conforming to MIL-H-5606 for a period of 24 hours. The tool shall be rejected if any evidence of deterioration or distortion results.

4.5.1 Performance test. - Each sample handling tool that passes the oil resistance test shall be cleaned and dried, and attached, by means of the vacuum cups, to a flat stationary surface. The surface shall be clean, dry, flat, smooth, and large enough to accommodate both cups. A pull of 120 pounds shall then be applied to the handle, perpendicular to the flat surface, and held for a period of 5 minutes. The tool shall be rejected if it pulls loose from the surface.

4.6 Inspection of preparation for delivery requirements. - An inspection shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table I. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

Table I. - Classification of Preparation for Delivery Defects.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102).

5.1 Packaging. - Packaging shall be level A, B, or C as specified (see 6.1).

5.1.1 Level A. - Each lifter shall be packaged in a close-fitting box conforming to MIL-B-43014. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 Level B. - Each lifter shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665 or PPP-B-676. The box shall be closed in accordance with the appendix to the box specification.

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5.1.3 Level C. - The lifters shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. - Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. - Lifters of like description packaged as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-585, class 3, PPP-B-591, class 2, PPP-B-601, type II, class 2, PPP-B-621, class 2, PPP-B-636, class 2 or PPP-B-640, class 2, grade A. The fiberboard boxes shall be closed in accordance with the appendix to the box specification. The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed approximately 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification. Strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. - Twelve lifters of like description packaged as specified in 5.1 shall be packed together in a close-fitting box conforming to PPP-B-636, class 1, style RSC. The box shall be closed with two strips of tape conforming to PPP-T-76 on top and bottom seams.

5.2.3 Level C. - The lifters shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. - In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. - In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packing and packaging required (see 5.1 and 5.2).
(c) Any special marking required (see 5.3).

Notice. - SEE LAST PAGE.

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ADDENDUM

INSTRUCTIONS ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards and Handbooks from established distribution points in their agencies.)

NOTICE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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